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Distributed IoT Communications With LEO Satellites: QoS Performance and Terminal Attempt Rate Schemes

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ABSTRACT Low Earth Orbit (LEO) satellites are increasingly being used to provide connectivity for wide area Internet of Things (IoT) sensing applications. Distributed IoT terminals are not able to coordinate their uplink transmissions to the nano-satellites, and so operate in a grant-free mode. Quality-of-Service (QoS) depends on the transmission attempt success statistics, which are time-varying. This paper presents asymptotic analysis results that characterize the IoT terminal's transmission success process. We show that it converges to an inhomogeneous Poisson process, in the large population regime, and characterize the time-dependent intensity as a function of the terminal location and the attempt rate scheme. We also propose three terminal attempt rate schemes that are solutions to max-min optimization problems. The performance of the proposed schemes are compared in terms of individual terminal QoS as well as the population-wide QoS distribution. Performance for various IoT applications is also presented.

INDEX TERMS LEO satellite communications, Internet of Things, direct-to-satellite IoT, massive access, many-access channel, grant free random access, multi-user interference cancellation.

I. INTRODUCTION

LOW EARTH Orbit (LEO) satellites offer an ideal communications solution for wide area Internet of Things (IoT) sensing applications [1], [2], [3], [4]. They provide a large coverage area, compared to terrestrial wireless networks, and do so with relatively low transmission power and lower latency, compared to geostationary satellites.

In this paper we focus on nano-satellites and IoT sensing applications that have low rate delay tolerant communication requirements on the uplink [2], and the terminals typically have limited processing capabilities and are extremely power constrained. The nano-satellites are also power and computation constrained, and are limited in number, with corresponding temporally intermittent coverage as they orbit overhead.[1] As such, there is a limited downlink channel, and uplink packets are transmitted in a *grant free* random access mode without packet acknowledgements due to the large number of terminals and small packet sizes [5], [6].

Ground terminals can only send packets when a LEO satellite is overhead. In some IoT applications, each transmission attempt sends a new data packet, while in other applications, a single data packet is attempted multiple times. For example, in an environmental monitoring sensing application, a terminal may have only a single message to send during a satellite pass (*e.g.,* a soil moisture monitor). Since there are no packet acknowledgements, the terminal will not be able to determine whether a packet is successful, and therefore it will send the same packet multiple times (in multiple transmission attempts) [7]. In multi-packet applications, the packets are not repeated, and instead, packet erasures are accommodated by coding across packets [8]. In each case, quality-of-service (QoS) depends on the transmission attempt success statistics, which are time-varying due to the terminal distribution and the LEO satellite flight path.

[1]Examples of current LEO nano-satellite based IoT sensing systems are the offerings of Myriota and Fleet Space Technologies.

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To mitigate erasures, multi-user interference cancellation[2] is used in LEO IoT systems to jointly detect packets [12], [13], [14] and this is a key determinant of transmission attempt success statistics and overall system throughput performance. Compressive sensing techniques and spreading codes have been used for activity detection [15], [16], [17], since active terminals are only a small fraction of the total population. Multi-user detection is then used for jointly detecting active terminals' packets (*e.g.,* using soft interference cancellation) [18], [19], [20]. These two tasks have been considered jointly in [13], [14], [15], [17], [21], [22]. In this paper, we propose a statistical model for multi-user detector performance and use it to address the crucial challenge of providing QoS guarantees to IoT terminals.

Packet success analysis for LEO IoT systems has previously been considered using a stochastic geometry approach [23], [24]. In [23], grant-free random access was considered for multiple satellites with overlapping footprints. The satellite locations were modelled by a random binomial point process on a sphere, and it assumed that all the ground terminals transmitted packets with equal probability. The assumption of random satellite locations was also used in [24], along with a Poisson cluster process to model IoT terminal locations. The packet success probability was derived for the case of grant-based access in which only a single user was scheduled in each satellite footprint.

In practice, LEO satellite orbits are predictable, and can be precomputed or estimated from broadcast beacon signals or from Doppler shifts [25]. The snapshot modeling approach used in [23], [24], with random satellite locations, does not account for the mobility of LEO satellites. In this paper, we take the satellite orbit into account, and assume the IoT terminals know the satellite visibility windows when a satellite is above a given elevation (for example by using approaches in [26]), and optimise the terminal transmission packet attempt rates as a function of the satellite orbits.

Optimising terminal packet access has previously been considered for LEO IoT, for grant-based systems. In [27], a user scheduling problem was formulated taking into account packet sizes, channel conditions, and satellite visibility times. The problem was a Knapsack problem that is NP hard. In [28], fairness across service slices was taken into account, and a mixed integer programming problem was formulated which is computationally challenging. In [29], [30], user scheduling was optimised for LoRa to LEO satellite systems. In [29], a first come first served TDMA approach using transmission slides of varying time durations were used to schedule IoT transmissions to avoid collisions. In [30], a mixed integer linear program was used to obtain a theoretical upper bound on scheduler performance.

In this paper, we provide a framework for QoS optimization for a LEO IoT network for grant-free random access. We show that the attempt process and the success process of a terminal converge to inhomogeneous Poisson processes in the large population regime, using weak-limit theory [31]. We characterize the respective time-dependent Poisson intensities as functions of the terminal location and the attempt rate scheme. We design three terminal attempt rate schemes that incorporate the limited satellite feedback and terminal complexity. Each scheme is based on a solution to a max-min fair optimization, under different feedback and terminal information restrictions. As such, they have different implementation complexities.

We show that our asymptotic results apply even for modest network sizes (*e.g.,* 240 terminals) that are well within the scale of practical IoT networks, using a numerical example from a real-world scenario. We provide detailed comparisons of the proposed attempt schemes using numerical simulation, for a variety of IoT application scenarios. Of our three proposed attempt rate schemes, the Joint scheme (where terminals adapt their attempt rate according to the satellite feedback, while also considering their visibility durations) is shown to have the best overall performance in all application scenarios. It achieves the best QoS for short windowed terminals without significantly sacrificing on the QoS of long windowed terminals. As IoT sensing deployments grow in scale in the future, this advantage becomes increasingly important.

II. SYSTEM MODEL

In this section, we explain how terminals distributed in a large area experience different satellite visibility durations, and we model the time-varying traffic load as the satellite moves overhead. We then present the multi-user interference cancellation model and determine the rate of successful transmissions in any given time slot.

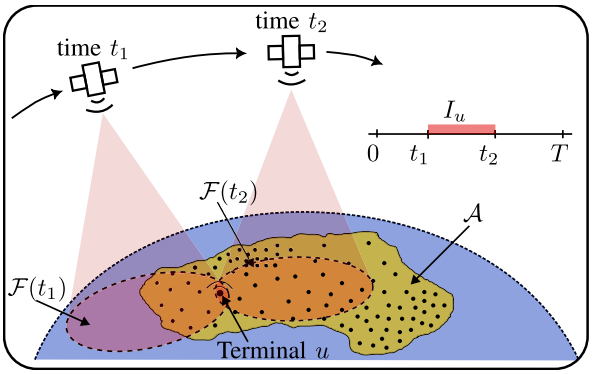
A. TERMINAL LOCATIONS AND SATELLITE VISIBILITY

Consider a period of time, starting at time 0 and ending at time T. We call this an Optimization Block (OB). This could be in the order of minutes, or many hours during which LEO satellites can orbit many times. Let $\mathcal{A}$ denote the geographical region that contains all the IoT terminals that are in the footprints of any of the system's satellites at any time between 0 and T. For example, consider a special case where all the customers of a LEO IoT service are on an island, as illustrated in Fig. 1(a). Alternatively, $\mathcal{A}$ could be the entire world. The terminal locations are fixed[3] and we denote the set of user terminal locations by $\mathcal{U}$.

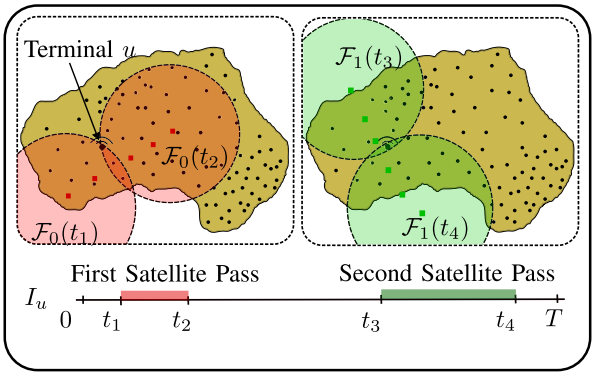
A satellite's footprint $\mathcal{F}(t) \subset \mathcal{A}$ is the set of locations (*i.e.,* geographical region) from which the satellite is visible above

[2]Multi-user detection was initially developed for terrestrial mobile communications [9], [10], [11].

[3]The assumption of fixed terminal locations does not imply that the terminals need to be stationary in practice. The large geographic scale of the satellite footprint means that unless a terminal moves extremely fast, its mobility will have little impact on system performance during one satellite pass. Moreover, the approach taken in this paper can be extended to handle highly mobile IoT nodes by allowing the terminal locations to vary from one optimization block to the next.



(a) Illustration of an example where a single satellite passes over the region $\mathcal{A}$. The satellite is visible to a terminal located at $u \in \mathcal{U}$ during the window $I_u = [t_1, t_2)$.



(b) An illustration of an example where a terminal $u \in \mathcal{U}$ is covered by two satellites during the OB, at two different times. The terminal's visibility window is $I_u = [t_1, t_2) \cup [t_3, t_4)$.

FIGURE 1. Illustration of Terminal Distribution and Satellite Pass.

the horizon (or above some minimum elevation). We assume that there are J nano-satellites and that their footprints, $\mathcal{F}_j(t)$, do not overlap at any time during $[0, T]$, which means that at any given time at most one satellite is visible to a terminal on the ground. As described in the Introduction, we are focused on IoT sensing services, where coverage is provided on a periodic basis from a small number of satellites (as opposed to a broadband network of thousands of satellites with overlapping footprints providing continual coverage). When a satellite passes over a terminal, there will either be another nano-satellite following behind it or another nano-satellite appearing on another orbit plane, and in single-satellite systems the next visible satellite will be the same satellite on its next pass.

The footprint of the satellite depicted in Fig. 1(a) is shown at times $t = t_1$ and $t = t_2$. Observe that terminal u is first covered by the footprint at time t_1 and last covered at time t_2. If this is the only satellite visible to the terminal during the optimization block (OB), we call the interval $[t_1, t_2)$ the terminal's visibility window, and denote it by I_u. If the OB is long enough that multiple satellites will pass over the terminal, then it will see the satellites at disjoint time intervals. In this case, we take the visibility window, I_u, to be the union of these time intervals, as illustrated in Fig. 1(b). In Fig. 1(b), the terminal at location u is in the footprint of the first nano-satellite during the interval $[t_1, t_2)$ and in the footprint of the second nano-satellite during the interval $[t_3, t_4)$. Thus, in this case, the visibility window for the terminal at u is given by $I_u = [t_1, t_2) \cup [t_3, t_4)$. Different terminals will be in a satellite footprint at different times (due to being at different locations) and will have different visibility windows.

Satellite orbits are deterministic, so IoT devices can wake up just prior to a satellite becoming visible, and go to sleep between satellites, in order to save power.

B. TERMINAL ATTEMPT FRAMEWORK

The terminal makes transmission attempts (using simple random access) to send its packets during the slots in which the satellite is visible (*i.e.,* the satellite is visible over the entire slot duration). Since LEO satellite orbits are deterministic, each terminal knows its own visibility window (since it knows where it is on the earth, via GPS for example, and it knows when the LEO satellites will pass overhead). The terminal's transmission attempts are made according to a terminal attempt rate function $\beta_u(t)$ that determines the attempt rate at time t. The terminal attempt rate function $\beta_u(t)$ can be viewed as a slow varying function that is fixed on the order of slot durations.

1) TERMINAL ATTEMPT RATE FUNCTIONS

We consider attempt rate functions which allow the terminal attempt rate to depend on the size of its visibility window, $|I_u|$, and it can be adapted over time based on a real-time signal, $w(t)$, that is broadcast from the satellite. The value of $w(t)$ is common to all terminals in the satellite footprint, and is broadcast from the satellite to all terminals in the footprint at time t.[4] This approach is chosen to reflect the practical restrictions on feedback in uncoordinated IoT multiple access channels and the limited computational capability of IoT sensors. Since the feedback is a common broadcast signal, and not particular to individual terminals, this approach scales to very large numbers of terminals, without any impact on the feedback rate.

The terminal attempt rate, $\beta_u(t)$, is defined to be a function of the terminal's visibility window and low rate feedback $w(t)$ sent from the satellite that is visible to the terminal at time t.

$$\beta_u(t) = h(|I_u|, w(t))\mathbf{1}_{I_u}(t) \tag{1}$$
$$\mathbf{1}_A(t) = 1, \ \ t \in A \subset [0, T], \ \ 0 \text{ otherwise}$$

[4]In practice, there will be discrete time instances when feedback is broadcast from the satellite, and $w(t)$ will be piece-wise constant changing at the feedback rate. In our numerical results, at the time when each slot starts, t, the value of $w(t)$ is broadcast by the satellite. Our results can also be generalized to vector valued feedback, *e.g.,* to deal with multiple terminal classes.

Note that the visibility window I_u is known locally at the terminals (from known satellite trajectories), whereas $w(t)$ is a global variable broadcast from the LEO satellite.

We now present the feedback requirements for three generic attempt rate schemes as follows:

- Uniform constant attempt rate scheme, where $h(|I_u|, w(t)) = h$ is the same constant function for all terminal locations and all times $t \in [0, T)$. This scheme requires no feedback as the constant rate can be apriori hardcoded into the terminals.
- Window length based attempt rate scheme, where $h(|I_u|, w(t)) = h(|I_u|, w)$ only depends on the window length of the terminal and stays fixed over time $t \in [0, T)$. This scheme also requires no feedback if the common scalar w is fixed, and is apriori hard-coded into the terminals. Typically, w will not change over a long time-scale (many OBs) in which case, the same scalar w can be broadcast to all terminals over this longer time period, using a low rate downlink channel.
- General attempt rate scheme, where $h(|I_u|, w(t))$ is changing over time as well as terminal locations. This scheme requires regular feedback from the satellite that is required to broadcast the value of $w(t)$ periodically, so that the terminals can set their rates accordingly.

In this paper, $w(t)$ will be restricted to piece-wise constant functions of time. Let T_f (in minutes) denote the interval length of periodic satellite feedback. Note then that $w(t)$ takes a constant value $w[i]$ during the interval $[iT_f, (i+1)T_f)$. The attempt rate function $h(|I_u|, w)$ is assumed to be bounded. This must hold in practice since terminal transmit power is heavily restricted.

2) TERMINAL TRANSMISSION ATTEMPT PROCESS

Time is divided into small slots of duration T_s minutes, where slot l corresponds to the interval $[lT_s, (l+1)T_s)$. Given satellite visibility for the entire slot $[lT_s, (l+1)T_s)$, a terminal at location u attempts a packet transmission with probability $p_u[l]$, defined as follows:

$$p_u[l] = \int_{lT_s}^{(l+1)T_s} \beta_u(\tau) d\tau \tag{2}$$

If the satellite is not visible during the entire slot, then $p_u[l] = 0$. We also assume that the transmission attempts are independent across terminals, making the transmission process an uncoordinated random access policy. Therefore the attempt process $A_u[l]$ of terminal u is a sequence of independent Bernoulli random variables, where the lth Bernoulli random variable takes value 1 with probability $p_u[l]$ and is 0 otherwise.

C. MULTI-USER INTERFERENCE CANCELLATION MODEL AND TRANSMISSION SUCCESS

We consider satellites that have the ability to decode multiple packets that arrive simultaneously from different users. This is modelled as follows: If $\overline{K}$ packets or fewer collide in a time slot, then there is a chance that some of those packets will be successfully decoded, with a uniform probability of success for each packet. However, if more than $\overline{K}$ packets collide, then none of them will be successful.

In this subsection, we consider a single satellite at an arbitrary time.[5] We denote the set of IoT terminals in the satellite footprint by $\mathcal{U}_f \subseteq \mathcal{U}$. Let S_k be the expected number of successful packets given that k of these terminals are attempting in slot l. A terminal at u makes a successful transmission in slot l with probability $\frac{S_k}{k}$, if

- The terminal at location u attempts a transmission in slot l, and
- The aggregate number of transmission attempts $A[l] = k$ from the footprint population in slot l.

We define the congestion process for u, $C_u[l] := A[l] - A_u[l]$, to be the total number of attempts from the terminals in $\mathcal{U}_f$ except for u, (defined as $\mathcal{U}_{\text{other}} := \mathcal{U}_f - \{u\}$), in slot l. The success probability of a packet, $\sigma_u[l]$, conditioned on terminal u's transmission in slot l, is therefore given by

$$\sigma_u[l] := \sum_{k=1}^{\overline{K}} \mathbb{P}[C_u[l] = k-1] \frac{S_k}{k} \tag{3}$$

Remark 1: Note that the conditional success probability $\sigma_u[l]$ is a function of the locations $\mathcal{U}_{\text{other}}$ of other terminals in the satellite footprint, since they are the ones that cause congestion to u's attempts. The conditional success probability is independent of the terminal's location, but it depends on the locations of the other terminals. Nevertheless, we index the conditional success probability with u (thinking of u as the identity of the terminal) since the set $\mathcal{U}_{\text{other}}$ is different for each terminal u.

By independence, the packet success process $S_u[l]$ is an independent Bernoulli random variable that takes value 1 with probability $p_u[l]\sigma_u[l]$.

The total number of successful packets aggregated across the footprint population $\mathcal{U}_f$ is now given by

$$S[l] := \sum_{u \in \mathcal{U}_f} S_u[l] \tag{4}$$

In [32], results on $S_u[l], A_u[l], S[l]$ were presented for a uniform constant attempt rate scheme where terminals have a single message for transmission. Our focus here is on the performance of more general attempt rate schemes as well as more general terminal applications, in the asymptotic regime with a large number of terminals.

To develop the framework for the asymptotic regime, the following continuous time representations of the processes are required. In continuous time, the processes are defined to be piece-wise constant (*i.e.,* remains fixed during the slot interval) as follows:

[5]Since the footprints of satellites do not overlap, multi-user cancellation is performed on a per-satellite basis.

$$\forall l \in \mathbb{N}, \forall t \in [lT_s, (l+1)T_s),$$
$$A_u(t) \coloneqq A_u[l], \quad C_u(t) \coloneqq C_u[l], \quad \sigma_u(t) \coloneqq \sigma_u[l],$$
$$S_u(t) \coloneqq S_u[l], \quad S(t) \coloneqq S[l], \tag{5}$$

The attempt process $A_u(t)$, the success processes $\sigma_u(t), S_u(t), S(t)$ and the congestion process $C_u(t)$ are all random functions, each depending on the random terminal locations and their Bernoulli attempts. To obtain the values of these quantities in a practical scenario requires Monte-Carlo simulation. The complexity of this approach scales linearly with the number of terminals and only provides limited insight into the network operation.

We take an alternate approach to characterize the large scale network dynamics using weak limit asymptotics. We will show that the conditional success probability $\sigma_u(t)$ converges to a deterministic function, independent of u, as the network scales large. Our framework provides greater insight into the bottlenecks of a large scale network. We will use it to derive results on optimal network performance, optimal attempt rate schemes and terminal QoS distribution.

III. ASYMPTOTIC PERFORMANCE CHARACTERIZATION

Consider a large-scale system in which the number of terminals $|\mathcal{U}|$ is very large, but the individual packets are very short, so that the slot time is very short. It follows that the number of time slots in the interval $[0, T]$ is very large. We consider a scaling in which the number of terminals, N, grows large, and the number of slots in $[0, T]$ also scales linearly with N. We will present convergence results that hold in the large-N limit. In this section, we consider the case where $J = 1$ *i.e.,* there is only one satellite in the system. This is just for notational simplicity; the results generalize immediately to multiple satellites.[6] The scaled networks indexed by N are considered as follows:

A. SCALED NETWORKS

In our model, the locations of terminals are independently and identically distributed (*i.i.d.*) in $\mathcal{A}$ according to a piece-wise continuous density $\nu(\cdot) : \mathcal{A} \to \mathbb{R}_+$. In other words, the probability that a terminal is located in a subset $A \subseteq \mathcal{A}$ is given by $\nu(A) \coloneqq \int_A \nu(u)du$. The density $\nu(\cdot)$ captures the statistics of terminal locations sufficient to characterize asymptotic performance. We also assume that the attempt rate function $h(|I_u|, w)$ is integrable with respect to the measure $\nu(du)$ for each w.

For the purposes of obtaining asymptotics, we construct a nested sequence of terminal locations $\mathcal{U}^{(N)}$ indexed by N (the population size) $\ldots \subseteq \mathcal{U}^{(N)} \subseteq \mathcal{U}^{(N+1)} \subseteq \mathcal{U}^{(N+2)} \subseteq \ldots$, starting from some initial set of *i.i.d* terminal locations. $\mathcal{U}^{(N+1)}$ is obtained from $\mathcal{U}^{(N)}$ by adding one more terminal location drawn independently, according to ν. We define $\nu^{(N)}$ to be the empirical distribution of terminal locations determined by $\mathcal{U}^{(N)}$.

[6]Each satellite can be treated independently under this section's framework given the assumption that their footprints do not overlap.

The slot size for the Nth network is

$$T_s^{(N)} \coloneqq \frac{1}{N\Gamma} \tag{6}$$

so that there are $N\Gamma$ slots in a minute. Hence, Γ is the slot rate per terminal, a bandwidth parameter that specifies the number of slots available per unit time, relative to the terminal population. In what follows, the processes for the Nth network are denoted by a superscript N, *e.g.,* $A_u^{(N)}(t)$ is the attempt process for a terminal at u in the Nth network.

To understand the large-N limit, consider a uniform constant terminal attempt rate scheme $\beta_u(t) = h$ attempts/minute, where the number of attempts $N\Gamma \int_{t_0}^{t_1} A_u^{(N)}(t)dt$ in a open interval $(t_0, t_1) \subset I_u$ is given by a Binomial random variable $B(\lfloor N\Gamma(t_1 - t_0)\rfloor, \frac{h}{N\Gamma})$ [32]. As $N \to \infty$, the number of slots $\lfloor N\Gamma(t_1 - t_0)\rfloor \to \infty$ and the per-slot attempt probability $\frac{h}{N\Gamma} \to 0$. In this limit, the Binomial distribution converges to a Poisson distribution (with mean $h \times (t_1 - t_0)$) (the so called *Poisson law of small numbers* [33]). Our asymptotic framework will be obtained along similar lines, where the (marginal) Poisson distribution and (time heterogeneous) Poisson processes are essential to characterizing the random properties of the network. The following terminology and notation will be used to present the main results.

For Poisson distributions, let $\mathcal{P}(n; \mu) \coloneqq e^{-\mu}\frac{\mu^n}{n!}$ denote the probability that a Poisson random variable with mean μ, takes the value $n \in \mathbb{Z}_+$. Using this notation, the *common success probability* function is defined as follows:

$$\Sigma(x) \coloneqq \sum_{k=1}^{\overline{K}} \mathcal{P}(k-1; x)\frac{S_k}{k}, \forall x \in \mathbb{R}_+ \tag{7}$$

Note that $\Sigma(x)$ is the transmission success probability for a transmitting terminal, if the overall number of transmissions is Poisson with mean x. $\Sigma(x)$ is similar in concept to $\sigma_u[l]$ in (3), and they are identical if $C_u[l]$ has a Poisson distribution with mean x. The precise relation between $\sigma_u^{(N)}(t)$ and $\Sigma(\cdot)$ will be presented in our Theorem 1 below.

Definition 1: A Poisson process with intensity function $r(t)$ is a random collection Π of points in $[0, T)$, such that

- the number of points $N(t_1, t_2)$ in an open interval (t_1, t_2) is distributed according to the Poisson distribution as $\mathbb{P}[N(t_1, t_2) = n] = \mathcal{P}(n; \int_{t_1}^{t_2} r(t)dt), \forall n \in \mathbb{Z}_+$, and
- $N(t_1, t_2)$ and $N(t_3, t_4)$ are independent for disjoint intervals (t_1, t_2) and (t_3, t_4).

For convenience, we will use the notation $\Pi_{r(t)}$ to denote a realization of the points of a Poisson process with intensity function $r(t)$ [34]. The Poisson process is homogeneous if $r(t)$ is a constant, and inhomogeneous otherwise.

B. ASYMPTOTIC REGIME RESULTS

Suppose $M \geq 1$ and choose a terminal $u \in \mathcal{U}^{(M)}$ in a realization of the Mth network. Using the per-slot attempt rate function in (1) and the definition for $\sigma_u(t)$ in (5), the success probability for u at time t in the given realization

of the Mth network can be obtained, namely $\sigma_u^{(M)}(t)$. Since the sequence of networks is nested, we have $u \in \mathcal{U}^{(N)}$ for all $N \geq M$, so the corresponding success probability for the same terminal u can be obtained for the Nth network, namely $\sigma_u^{(N)}(t)$. The following theorem gives the asymptotic limit of $\sigma_u^{(N)}(t)$ as the network grows large.

Definition 2: The expected number of attempts per slot made by the terminals in the footprint, is denoted $H^{(N)}(t)$ at time t, and defined as follows:

$$H^{(N)}(t) = \frac{1}{\Gamma} \int_{u \in \mathcal{F}(t)} h(|I_u|, w(t)) \nu^{(N)}(du) \tag{8}$$

where $\nu^{(N)}(du)$ is the empirical terminal distribution in the Nth network as mentioned earlier.

Theorem 1: As $N \uparrow \infty$, the following limit holds almost surely:

$$H^{(N)}(t) \to H(t) \text{ uniformly over } t \in [0, T] \tag{9}$$

where $H(t) := \frac{1}{\Gamma} \int_{u \in \mathcal{F}(t)} \nu(u) \beta_u(t) du$ is the asymptotic time dependent intensity function.

In addition, terminal conditional success probabilities converge to a common limit:

$$\sigma_u^{(N)}(t) \to \Sigma(H(t)) \text{ uniformly} \tag{10}$$

for each $u \in \mathcal{U}^{(N)}$, almost surely.

Proof: See Appendix A. ∎

As explained in Remark 1, the pre-limit success probability $\sigma_u^{(N)}(t)$ depends on the identity of terminal u, since the terminal's identity implicitly defines the physical locations of the other terminals in the given network realization. Intuitively for large N, the dependence of $\sigma_u^{(N)}(t)$ on the terminal's own identity will be very weak (and will vanish as $N \to \infty$), since removing any one point from a large set has negligible impact on overall statistical behaviour. Theorem 1 confirms this and also characterizes the asymptotic success probability as the function $\Sigma(H(t))$.

Theorem 2: As $N \uparrow \infty$, the attempt process and success process (defined in (5)) of a terminal $u \in \mathcal{U}^{(N)}$ converge to Poisson processes as follows:

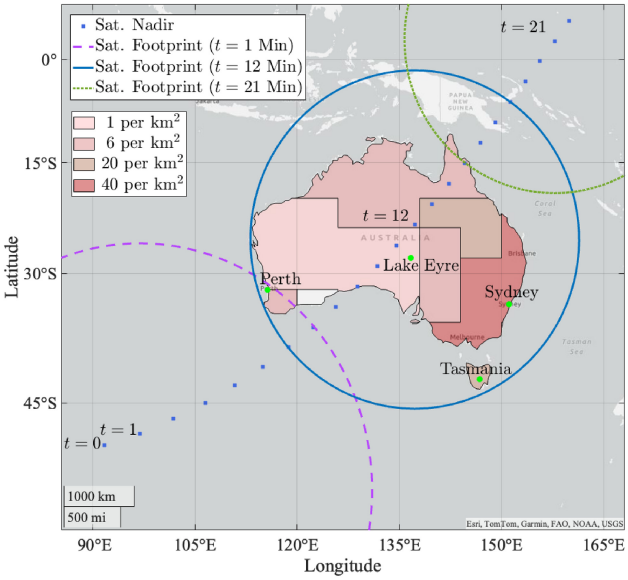
$$A_u^{(N)}(t) \Rightarrow \Pi_{\beta_u(t)} \tag{11}$$

$$S_u^{(N)}(t) \Rightarrow \Pi_{\beta_u(t)\Sigma(H(t))} \tag{12}$$

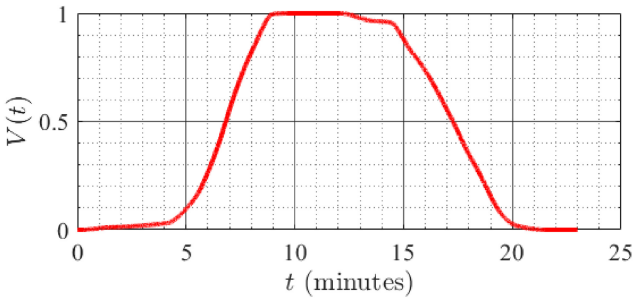
where $\Rightarrow$ denotes weak convergence.

Proof: See Appendix B. ∎

The main implications of Theorem 2 are that, asymptotically, the attempt process of terminal u is a Poisson process with rate function $\beta_u(t)$, and the success process is obtained by marking this process with independent successes, with success probability $\Sigma(H(t))$ at time t. The expected number of attempts and successes by terminal u during a time interval $\mathcal{T}$ are given by $\int_{t \in \mathcal{T}} \beta_u(t) dt$ and $\int_{t \in \mathcal{T}} \beta_u(t) \Sigma(H(t)) dt$, respectively. In general, both processes are inhomogeneous Poisson processes, although the attempt process is homogeneous on I_u if $\beta_u(t)$ is constant on I_u. It also follows from the theorem that the success processes



(a) Terminal distribution and the track of satellite nadir. The footprints are shown for three-time instances $t = 1, 12$ & 21 minutes.



(b) Fraction of population covered by satellite's footprint.

FIGURE 2. Terminal Distribution and Satellite Pass.

of any two terminals (or indeed any finite number) are independent in the asymptotic limit.[7]

C. CONTINENTAL SCALE NUMERICAL EXAMPLE

We now provide an illustration of Theorem 1 using a numerical example where IoT terminals are distributed over the continent of Australia. As Theorem 1 is an asymptotic, it is crucial to identify the scale (order of N) at which the limits take hold. Therefore, we present results for N ranging from 30 terminals to 480 terminals. It will be seen that our asymptotic results apply for modest network sizes of $N = 240$ and above, well within the scale of practical IoT networks.

The setup we consider is as follows. The terminals are positioned based on livestock density in Australia, and are placed at random locations *i.i.d.* according to the piecewise constant density $\nu(\cdot)$ shown in Fig. 2(a). The colour code over Australia presents the livestock density per square kilometre, derived from the statistics in [35]. We consider

[7]This of course does not imply that the ensemble of all success processes (which grows large) are independent.

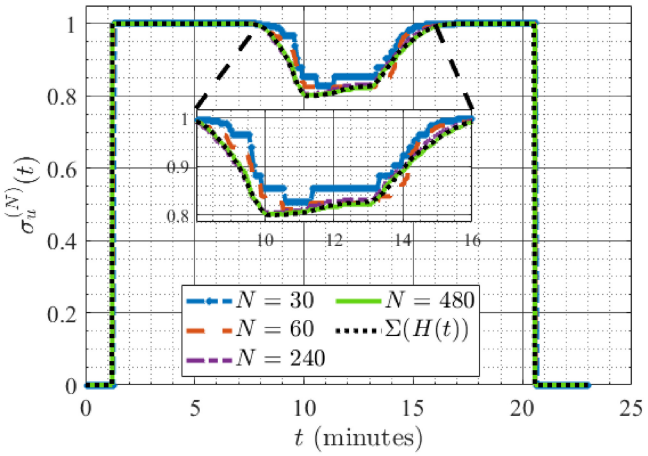


FIGURE 3. The success probability of a packet, $\sigma_u^{(N)}(t)$, for terminal u transmitting, approaches the Poisson limit $\Sigma(H(t))$ as N increases, for $\Gamma = 0.5$.

a satellite (at 500 kilometre altitude) flying over Australia in a Northeast direction. Fig. 2(a) shows the track of the satellite nadir (blue squares) at one-minute intervals, as well as the satellite footprints (circles) at times $t = 1, 12$, and 21 minutes. The total duration T is taken to be 23 minutes. This setup was also considered in [32] where a uniform constant attempt rate scheme was considered (however, the accuracy as a function of N was not investigated, as we do in this section).

We adopt the same uniform attempt rate scheme [32], whereby each terminal attempts at a rate of $\beta = 3.65$ attempts per minute during satellite visibility, and Γ is taken to be 0.5, which means that the slot duration is $2/N$ minutes, *e.g.,* 250 msec when $N = 480$. Hence, $H(t) = \frac{\beta}{\Gamma} V(t)$ where $V(t)$ is the fraction of terminal population in the footprint at time t. From $t \approx 9$ to $t \approx 12$ minutes, the satellite footprint fully covers Australia. Fig. 2(b) illustrates how the expected fraction of the terminal population varies as the satellite's footprint traverses Australia. Tasmania and Western Australia start to leave the footprint at $t \approx 12$ minutes (see Fig. 2(a)) and will spend less time inside the footprint than other parts of the country.

For multi-user cancellation, we take $S_k = k$ for $\overline{K} = 10$, which means perfect detection if there are ≤ 10 attempts in a slot, no packets are detected otherwise.

Fig. 3 illustrates the rate at which the success probability $\sigma_u^{(N)}(t)$ of a transmitted packet from terminal u converges to the proposed analytical limit $\Sigma(H(t))$ as N increases. In the figure, the black dotted curve depicts the asymptotic success probability $\Sigma(H(t))$, which is ≈ 1 at the beginning and end of the pass due to low congestion during these periods. Between $t \approx 8$ and $t \approx 16$ minutes, the satellite experiences a higher load as its footprint covers the densely populated southeast region (see Fig. 2(b)). Consequently, the packet success probability decreases and reaches a minimum of 0.7998 at $t \approx 10$ minutes. It can be observed that the success probability curve $\sigma_u^{(N)}(t)$, when $N = 480$, and $\Gamma = 0.5$ (*i.e.,* a slot duration of 250 msec), is already extremely close to the analytical limit $\Sigma(H(t))$. This shows that our asymptotic framework is quite accurate even for modestly sized IoT networks with realistic slot durations.

Clearly, for a fixed Γ, the slot duration tends to zero in the asymptotic limit, and it may seem that the slots are too short to send even a small number of bits when N is extremely large. However, in practice, the system bandwidth will determine the slot duration, and hence $N\Gamma$, and the terminal population determines N, which together determines the value of Γ and the corresponding limiting curve.

IV. OPTIMIZED ATTEMPT RATE SCHEMES

In this section, we will formulate max-min fair attempt rate optimizations to achieve load balancing across terminals in terms of their expected number of packet attempts. We will propose attempt rate schemes based on the solutions to these optimizations. These schemes will be used in Section V to obtain QoS results for various IoT applications. As in Section III, we will begin with the simplifying assumption that $J = 1$ (*i.e.,* one satellite in the system) but we will then extend the results to the general case with multiple satellites.

A. OPTIMAL NETWORK THROUGHPUT AND AGGREGATE ATTEMPT RATE

By Theorem 1, the mean number of attempts per slot is $H(t)$ and the attempt success probability is $\Sigma(H(t))$. Hence, the network throughput is $H(t)\Sigma(H(t))$ successes per slot. Let $\beta_{\max} := \arg\max_{x>0} x\Sigma(x)$, *i.e.,* $\beta_{\max}$ is the optimal value of $H(t)$ that achieves the highest network throughput. Interestingly, $\beta_{\max}$ may not be uniquely defined in general. Nevertheless, we will show that it is uniquely defined under an innocuous assumption on the multi-user canceller.

Recall that under the multi-user interference canceller, $\frac{S_k}{k}$ is the probability of transmission success of a terminal, given a total of k terminals transmitting simultaneously. We assume that this success probability $\frac{S_k}{k}$ decreases with increasing the number of transmitting terminals k. Under this reasonable assumption, the following lemma shows that $\beta_{\max}$ is unique.

Lemma 1: Suppose that $\frac{S_k}{k} \geq \frac{S_{k+1}}{k+1} > 0$ for each $k \geq 1$. Then the function $x\Sigma(x)$ has a unique stationary point in $(0, \infty)$.

Proof: See Appendix C. ■

Hence $\beta_{\max}$ is the unique value for aggregate per-slot attempt rate that maximizes the network throughput. Thus, for maximal network throughput performance, the aggregate per-slot attempt rate should be $\beta_{\max}$. A challenge then would be to allocate terminal attempt rates subject to this aggregate attempt rate constraint.

In this paper, we consider the challenge of maximizing the expected number of successful packet transmissions in the interval $[0, T]$ for the worst-case terminal, where the focus is on the QoS performance of individual terminals. (Of course, our methodology applies to other criteria, as well.) In other words, we consider max-min fair optimization, rather than total throughput optimization. In max-min fair optimization, there is no point in having the aggregate attempt rate higher

than the rate that maximizes the network throughput, and so $\Gamma\beta_{\max}$ appears as an upper bound constraint on the aggregate attempt rate, rather than an equality constraint. As mentioned before, the attempt rate allocation problem needs to consider the diversity of visibility window lengths across terminals and also the limited satellite feedback.

B. WINDOW BASED ATTEMPT RATE SCHEME

Consider first the scenario in which terminal u sets its attempt rate $\beta_u(t)$ to be $\beta_u \mathbf{1}_{I_u}(t)$, *i.e.,* each terminal chooses its own rate but fixed over its visibility window. We formulate a max-min fair optimization problem to maximize the expected number of attempts w of the worst-case terminal in the network as follows:

$$\max_{\beta_u \geq 0} \; w \tag{13}$$

$$\text{s.t.} \int_{t\in I_u} \beta_u \; dt \geq w, \;\; \forall u \in \mathcal{A} \tag{14}$$

$$\frac{1}{\Gamma} \int_{u\in\mathcal{F}(t)} \beta_u \nu(u) du \leq \beta_{\max}, \;\; \forall t \in [0,T] \tag{15}$$

Constraint (14) ensures that the expected number of attempts of each terminal u is greater than the objective value w. Constraint (15) requires that the aggregate per-slot attempt rate of the population to be below $\beta_{\max}$ at each time instant t. Theorem 3 shows that the max-min fair attempt rate scheme in this case is a window length based strategy: $\beta_u = h(|I_u|, w)$.

Theorem 3: An optimal max-min fair attempt rate allocation $\boldsymbol{\beta}^*$, *i.e.,* an optimal solution of LP (13)-(15) is given by

$$\beta_u^* := \frac{w^*}{|I_u|} \tag{16}$$

where $w^* := \frac{\Gamma\beta_{\max}}{\int_{u\in\mathcal{F}(t^*)} \frac{\nu(u)}{|I_u|} du}$ is the optimal objective value, and $t^* := \arg\max_{t\in[0,T]} \int_{u\in\mathcal{F}(t)} \frac{\nu(u)}{|I_u|} du$.

Proof: See Appendix C. ■

Note that with $\boldsymbol{\beta}^*$ satisfying (16), the maximum value of the LHS of (15) is achieved at the time t^* given in the theorem, and at time t^*, constraint (15) is met with equality. It follows that t^* is a bottleneck time instant at which the aggregate attempt rate is at a maximum, which is the throughput-maximizing aggregate attempt rate. The aggregate attempt rate can be *strictly less* than this maximum attempt rate at all other times, which means that the total throughput is not maximized. In general, maximizing the worst-case success rate of individual terminals does not maximize the total throughput.

Our optimization approach in (13)-(15) can be generalized to a constellation of $J > 1$ satellites (indexed by $j = 0, \ldots, J-1$) as follows. Let $\mathcal{F}_j(t)$ denote footprint of satellite j at time t. For each satellite there will be a constraint corresponding to (15), where in the jth constraint, $\mathcal{F}(t)$ is replaced with $\mathcal{F}_j(t)$. The optimal solution will again be of the form $\beta_u^* := w^*/|I_u|$ (as in Theorem 3), where

$$w^* = \frac{\Gamma\beta_{\max}}{\int_{u\in\mathcal{F}_{j^*}\left(t_{j^*}^*\right)} \frac{\nu(u)}{|I_u|} du}, \tag{17}$$

where $j^* := \arg\max_{j\in\{0,1,\ldots J\}} \max_{t\in[0,T]} \int_{u\in\mathcal{F}_j(t)} \frac{\nu(u)}{|I_u|} du$ and $t_j^* := \arg\max_{t\in[0,T]} \int_{u\in\mathcal{F}_j(t)} \frac{\nu(u)}{|I_u|} du$.

Theorem 3 shows that when a fixed attempt rate is adopted based only on the terminal location, the optimal policy is to set the attempt rate of each terminal inversely proportional to the length of its visibility window.[8] Note that this optimal solution lines up with the intuition of providing higher rates to terminals with shorter visibility windows, as a means of improving performance. Moreover, this attempt policy is of the form $\beta_u = h(|I_u|, w)$, and hence it has a low feedback implementation as discussed in Section II-B.

For this Window scheme, w^* will be computed for each OB by the service provider in a central processor (*e.g.,* in the Cloud). It will have knowledge of the terminal location density, ν, and the values of the visibility windows of the terminals, which can be calculated every OB based on knowledge of the satellite orbits for that OB (which is deterministic). The integrals in the definition of w^* are in practice discrete sums, of very low complexity, since the terminal density function, $\nu(\cdot)$, will be piece-wise constant in practice. To compute the sum, the value of t_j^* is required, which simply requires a one-dimensional search for each satellite, once per OB. Every OB, the value of w^* needs to be recomputed by the service provider and sent to the satellites. The satellites broadcast w^* in every timeslot to ensure that all terminals have the value at the start of their respective visibility windows.

In practice, for a scheme of this form (with a fixed w that does not change across the time-slots in an OB) it makes sense to keep w fixed for many OBs, greatly relaxing the downlink feedback rate to the terminals compared to schemes where $w(t)$ is a function of t. Using such an approach will only be an *approximation* of the Window scheme, because the optimization problem is not solved in each OB, but only relatively infrequently.

C. FOOTPRINT LOADING BASED ATTEMPT RATE SCHEME

Consider now a different scenario in which all the terminals in the satellite footprint adopt a common attempt rate β_t at time t. We formulate a max-min fair optimization problem for this scenario as follows.

$$\max_{\beta_t \geq 0} \; w \tag{18}$$

$$\text{s.t.} \int_{t\in I_u} \beta_t dt \geq w, \;\; \forall u \in \mathcal{A} \tag{19}$$

[8]Note that the attempt rate is limited by the terminal transmit power and cannot be arbitrarily scaled up for very short visibility windows. This issue is simply avoided by having a minimum window duration for a terminal to be considered as part of the optimization.

$$\frac{1}{\Gamma}\int_{u\in\mathcal{F}(t)}\beta_t \nu(u)du \le \beta_{\max},\ \ \forall t\in[0,T] \quad (20)$$

This optimization also has a closed form solution given by the following theorem.

Theorem 4: An optimal attempt rate allocation $\boldsymbol{\beta}^*$ to optimize the worst case expected number of attempts, *i.e.,* an optimal solution of LP (18)-(20), is given by

$$\beta_t^* := \frac{\Gamma\beta_{\max}}{V(t)} \quad (21)$$

where $V(t) = \int_{u\in\mathcal{F}(t)}\nu(u)du$ is the expected fraction of terminal population inside the footprint at time t.

Proof: See Appendix C. ■

Our optimization approach in (18)-(20) can be generalized to a constellation of $J > 1$ satellites (indexed by $j = 0,\ldots,J-1$), as in the previous subsection. For this case, all the terminals in the footprint of satellite j adopt a common attempt rate of $\beta_{j,t}$ at time t. There will be J aggregate attempt constraints in (20) where $\mathcal{F}(t)$ is replaced with $\mathcal{F}_j(t)$ and β_t replaced with $\beta_{j,t}$ for $j = 0,\ldots,J-1$. The optimal solution will be of the form $\beta_{j,t}^* := \Gamma\beta_{\max}/V_j(t)$ where $V_j(t) := \int_{u\in\mathcal{F}_j(t)}\nu(u)du$ is the fraction of the terminal population in the footprint of satellite j at time t.

Theorem 4 shows that when a common attempt rate is adopted at each time instant, but is allowed to vary across time, the optimal policy achieves the throughput-maximizing aggregate attempt rate at all times. The optimal scheme can be considered as a processor sharing type scheme, that shares the throughput maximizing aggregate rate $\Gamma\beta_{\max}$ (in attempts/slot) equally among the active terminal population at each time t. Also note that this attempt policy is of form $\beta_u(t) = h(|I_u|, w(t))$, and hence it requires periodic satellite feedback as discussed in Section II-B.

As with the Window scheme in the previous subsection, in this Footprint scheme the values of $V_j(t)$ can be computed by a central processor (*e.g.,* in the Cloud). It will have knowledge of the spatial intensity ν, and the durations of the visibility windows of the terminals, which can be calculated every OB based on knowledge of the satellite orbits during that OB (which is deterministic). In practical implementations, the integral in the definition of $V_j(t)$ will be a discrete sum, of very low complexity, since the terminal density function will be piece-wise constant in practice. These calculations are of the same complexity as the computation of the integrals in the Window scheme, but a one-dimensional search for the maximum value is not required. For each satellite, the function $V_j(t)$ needs to be sent up to the satellite, which then broadcasts the values in every time-slot t over the duration of the OB.

D. JOINT WINDOW-FOOTPRINT ATTEMPT RATE SCHEME

Although the Footprint scheme makes full use of the network capacity at each time instant, the Window scheme is superior in terms of load-balancing between terminals of different window lengths. The Window scheme generally achieves a higher max-min expected number of attempts. Hence we now construct a joint scheme that exploits the best properties of both the schemes.[9] Under this, the rate function for terminal location u (that is in the footprint of satellite j at time t) is set as

$$\beta_u(t) := \frac{z_j(t)}{|I_u|} \quad (22)$$

where

$$z_j(t) := \frac{\Gamma\beta_{\max}}{\int_{u\in\mathcal{F}_j(t)}\frac{1}{|I_u|}\nu(u)du} \quad (23)$$

for $j = 0,\ldots,J-1$. Note that this policy is a hybrid between the two preceding optimization solutions. On the one hand, the full aggregate attempt rate $\Gamma\beta_{\max}$ is exploited at each time instant as with the Footprint scheme, whereas on the other hand, the attempt rates are weighted across users according to the visibility windows as with the Window scheme. The computational complexity and amount of satellite feedback required (both uplink from the central processor and downlink from the satellite to the terminals) is the same as for the Footprint scheme.

Let w^* be the optimal objective of window based optimization. Its clear that under the Joint scheme, $z_j(t) \ge w^*$ for each j,t. Hence the attempt rate of each terminal under the Joint scheme is greater than or equal to the Window scheme, for all times t. Therefore, the max-min expected number of attempts is also greater under the Joint scheme.

E. COMPARISON FOR A TWO CLUSTER NETWORK

We now evaluate the three schemes using a numerical example to illustrate the main differences. We also derive a new scheme (for the two-cluster network) that utilises all the global information including actual visibility intervals and terminal locations. We call this the Global Information Scheme (GIS), and we prove that it is optimal in a max-min sense. As such, it provides an absolute upper bound on the other three schemes for the two-cluster network.

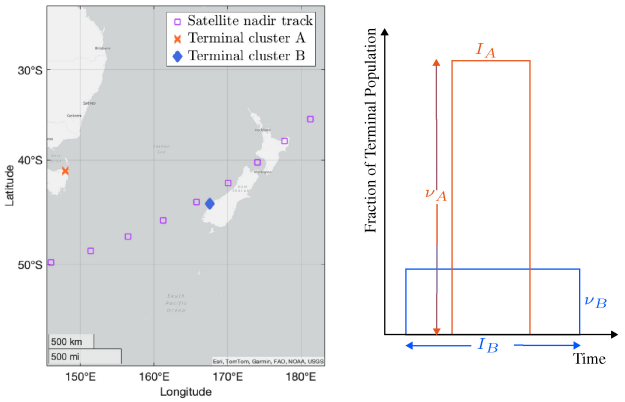
Consider a two-cluster network, with clusters A and B clusters A and B, of co-located terminals as shown in Fig. 4(a). The fraction of terminals are ν_A and ν_B (where $\nu_A + \nu_B = 1$), and the visibility windows are denoted I_A and I_B as shown in Figure. 4(b). With no loss of generality, we assume $|I_B| > |I_A|$. We define the population ratio $\eta := \frac{\nu_A}{\nu_B} \in (0,\infty)$ and the window ratio $\alpha := \frac{|I_A|}{|I_B|} \in (0,1)$.

The following LP maximizes the worst-case expected number of attempts under the assumption of global information for the two cluster scenario. Here, $\beta_B^{(c)}, \beta_A$ are the attempt rates of a cluster B terminal and a cluster A terminal respectively, during the congestion interval when

[9] A max-min fair optimization involving a general attempt rate variable $\beta_{u,t}$ would not be tractable, as no closed form solutions exist, and a numerical approach is prohibited by the large number of terminal locations. However, the full optimization for a special case of two terminal clusters will be presented in Section IV-E for a performance comparison with the proposed schemes.

TABLE 1. Attempt rates for the four schemes.

	β_A	$\beta_B^{(c)}$	$\beta_B^{(nc)}$
GIS	$\Gamma\beta_{\max}\min\left(\frac{\eta+1}{\eta}, \frac{1}{\alpha}\right)$	$\Gamma\beta_{\max}\max\left(0, \eta+1-\frac{\eta}{\alpha}\right)$	$\Gamma\beta_{\max}(\eta+1)$
Joint	$\Gamma\beta_{\max}\left(\frac{\eta+1}{\eta+\alpha}\right)$	$\Gamma\beta_{\max}\left(\frac{\alpha(\eta+1)}{\eta+\alpha}\right)$	$\Gamma\beta_{\max}(\eta+1)$
Window	$\Gamma\beta_{\max}\left(\frac{\eta+1}{\eta+\alpha}\right)$	$\Gamma\beta_{\max}\left(\frac{\alpha(\eta+1)}{\eta+\alpha}\right)$	$\Gamma\beta_{\max}\left(\frac{\alpha(\eta+1)}{\eta+\alpha}\right)$
Footprint	$\Gamma\beta_{\max}$	$\Gamma\beta_{\max}$	$\Gamma\beta_{\max}(\eta+1)$



(a) Satellite Pass (b) Visibility Windows

FIGURE 4. Two Cluster Example for a Single Satellite Pass. Terminals in cluster B experience a longer visibility duration ($|I_B| > |I_A|$) since the satellite passes directly overhead cluster B.

both clusters are in the footprint. $\beta_B^{(nc)}$ is the attempt rate of a cluster B terminal during the non-congestion period, when only cluster B is in the footprint (and not cluster A).

$$\text{maximize } w \text{ subject to} \tag{24}$$

$$\frac{1}{\Gamma}\left(\nu_A\beta_A + \nu_B\beta_B^{(c)}\right) = \beta_{\max};\ \ \frac{1}{\Gamma}\nu_B\beta_B^{(nc)} = \beta_{\max} \tag{25}$$

$$\beta_B^{(c)}|I_A| + \beta_B^{(nc)}(|I_B| - |I_A|) \geq w;\ \ \beta_A|I_A| \geq w \tag{26}$$

Constraint (25) ensures that the aggregate per-slot attempt rate is always at the maximum value $\beta_{\max}$. Constraint (26) ensures that the expected number of attempts for all the terminals is at least w. The solution to the above LP provides the optimal attempt rates under global information, and we call this the Global Information Scheme (GIS).

Theorem 5: For the two cluster network, the optimal attempt rates of the GIS scheme (*i.e.,* the solution to LP (24)-(26)) are given in the first row of Table 1. The corresponding optimal throughputs are given in the first row of Table 2. The attempt rates of the Joint, Window and Footprint schemes for the two-cluster network are also given in Table 1, and the corresponding throughputs are given in Table 2.

Proof: See Appendix D. ■

Consider the trivial case of $I_A = I_B$, where both clusters (and hence all terminals) are equivalent. In this case, the expected throughput[10] is the same for all the terminal, under all of the four schemes, and is given by $\Gamma\beta_{\max}\Sigma(\beta_{\max})|I_B| =: \overline{R}$. For the general case, the expected throughputs are given in Table 2, presented relative to $\overline{R}$.

[10]The expected number of successful transmissions over the time interval $[0, T]$.

TABLE 2. Expected throughputs under the four schemes.

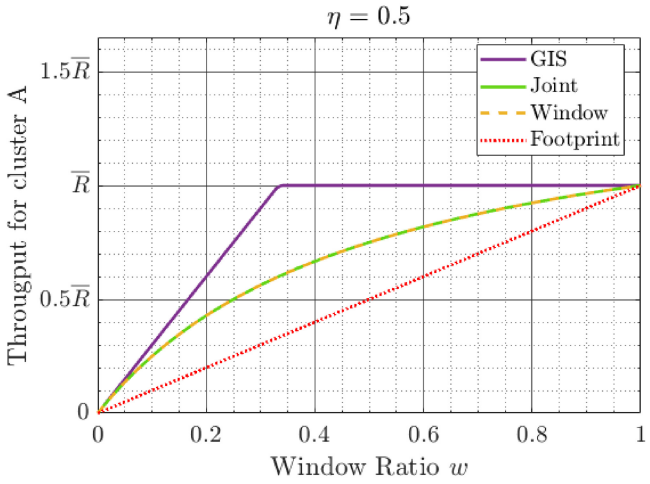
	Terminal Throughput	
	Cluster A	Cluster B
GIS	$\overline{R}\min\left(\frac{\alpha(\eta+1)}{\eta}, 1\right)$	$\overline{R}\max\left((1-\alpha)(\eta+1), 1\right)$
Joint	$\overline{R}\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)$	$\overline{R}(\eta+1)(1-\frac{\eta\alpha}{\eta+\alpha})$
Window	$\overline{R}\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)$	$\overline{R}\alpha\left(\frac{\eta+1}{\eta+\alpha}\right)\left(\alpha+(1-\alpha)\frac{\Sigma\left(\frac{\alpha\beta_{\max}}{\eta+\alpha}\right)}{\Sigma(\beta_{\max})}\right)$
Footprint	$\overline{R}\alpha$	$\overline{R}(\eta+1-\alpha\eta)$

Observe from Table 2 that when $\alpha > \frac{\eta}{\eta+1}$, all terminals achieve a common expected throughput of $\overline{R}$, for the GIS scheme. Also note from Table 1 that $\beta_B^{(c)} = 0$ when $\alpha < \frac{\eta}{\eta+1}$, which means that, in this scenario, only terminals in cluster A make attempts during the time interval I_A, and gives them the highest possible throughput of $\overline{R}\alpha\left(\frac{\eta+1}{\eta}\right)$. Thus, the GIS scheme achieves max-min fairness of expected throughputs.

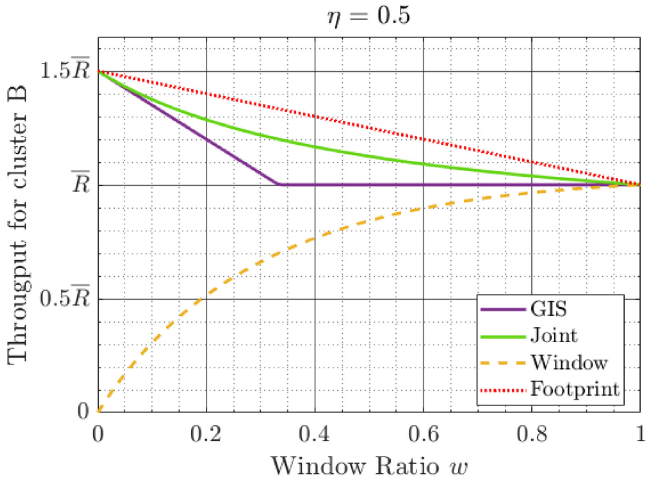
For the Window scheme, observe from Table 1 that the attempt rates of terminals in cluster B are constant across the whole time interval I_B, which is consistent with this scheme not requiring time-varying feedback from the satellite. In contrast, the attempt rates of terminals in cluster B change during time interval I_B for both the Footprint and Joint schemes. However, as shown in Sections IV-B and IV-C, respectively, they require only scalar time-varying feedback from the satellite (in general).

Table 2 shows that neither the Window scheme, nor the Joint scheme, achieve max-min fairness of expected throughputs across the terminals. The max-min fair optimization problems in (13)-(15) and (18)-(20) concern fairness in *attempt opportunities*, not fairness in expected throughput. To achieve max-min fairness of expected throughputs requires global information about the visibility windows of all terminals, as used in the optimization problem (24)-(26). It is not practical to obtain this information in the general case with a large number of locations.

Note that since $|I_A| \leq |I_B|$, the minimum throughput value is always achieved by the terminals in cluster A, for all schemes. Figure. 5(a) shows the cluster A throughput for each of the schemes as a function of window ratio α, for $\eta = 0.5$. It can be observed that the cluster A throughput under the Window and Joint schemes is greater than under the Footprint scheme, but lower than under the GIS scheme. The cluster A throughput of the GIS scheme is at the most 40%



(a) Expected throughput for a Cluster A terminal.



(b) Expected throughput for a Cluster B terminal.

FIGURE 5. Comparison of schemes in the two cluster scenario.

higher than the Joint scheme, which occurs when $\alpha = 1/3$. The cluster A throughput of the GIS scheme is 65% higher than the Footprint scheme when $\alpha = 1/3$.

Figure. 5(b) shows the cluster B throughput of each of the schemes as a function of window ratio α, for $\eta = 0.5$. Note that the Window scheme has the lowest cluster B throughput, especially for small window ratios. Under the Window scheme, the attempt rate is inversely proportional to the window size. Hence, the attempt rate assigned to a terminal in cluster B is α times the attempt rate of a terminal in cluster A, which is small when α is small. This is a good strategy during the congestion period where both clusters A and B are in the satellite footprint, since the link capacity is then predominantly dedicated to the terminals with small visibility windows. However during non-congestion when only cluster B is active, the aggregate attempt rate is significantly below the optimal value $\Gamma\beta_{\max}$, which leads to poor resource usage.

The Joint scheme overcomes this problem by scaling the attempt rates to meet the optimal value $\Gamma\beta_{\max}$, and hence achieves much higher throughput for cluster B terminals, as can be seen in Figure. 5(b). As can be seen, the throughputs of cluster B terminals in the Footprint and Joint schemes are both higher than in the GIS scheme, but this is achieved at the cost of lower throughput for cluster A which contains the max-min terminals.

Although the GIS scheme provides higher max-min throughput, it has no closed form solution. The computational complexity increases linearly with the number of cluster locations, and hence is impractical for networks with numerous wide spread terminal locations. Furthermore, the attempt rate function $\beta_u(t)$ would not be of the form $h(|I_u|, w(t))$. In fact, computation of $\beta_u(t)$ in the GIS scheme requires global information about the visibility windows of all the terminals, as can be seen from our two cluster example, which requires either the computations to take place non-locally (*i.e.,* in the cloud) and then broadcast to the terminals, or else the information for the computation must be distributed amongst all the terminals. Either way, this is a large overhead[11] which is avoided by the schemes based on the form $h(|I_u|, w(t))$ that we propose in this paper.

This numerical example demonstrates that the proposed attempt rate schemes (Window, Footprint and Joint) have generally good performance using minimal satellite feedback and local visibility window length information. The throughput of the Joint scheme is generally good, better than the Window scheme, and does not require onerous feedback from the satellite, although it requires more feedback than the Window scheme. The Window scheme is good in some scenarios (*e.g.,* higher window ratio α). These numerical results also illustrate the potential gain from more complex schemes that benefit from more information at the terminals, albeit requiring feedback of that information from the satellite, clearly an onerous requirement when the number of terminals is large.

V. NUMERICAL RESULTS FOR IOT APPLICATIONS

This section provides a performance comparison of the proposed attempt rate schemes, for various IoT applications. The comparisons here include network wide as well as terminal specific performance metrics.

A. SIMULATION SETUP

We consider a network with widely distributed terminals using the same experimental setup as detailed in Section III-C. See Fig. 2(a) for the terminal distribution $\nu(\cdot)$ and the satellite track details. Recall that for the multi-user interference cancellation model, we take $S_k = k$, $\forall k \leq \overline{K}$, *i.e.,* all the transmissions in a slot are successful provided that at most $\overline{K}$ terminals are attempting.

We compare the performance of the four attempt rate schemes: Uniform [32], Window, Footprint, and Joint schemes. The Uniform scheme uses a common attempt

[11]The overhead is not significant for the simple two cluster example considered in this paper, but would be very significant in real-world applications with very large numbers of terminal locations.

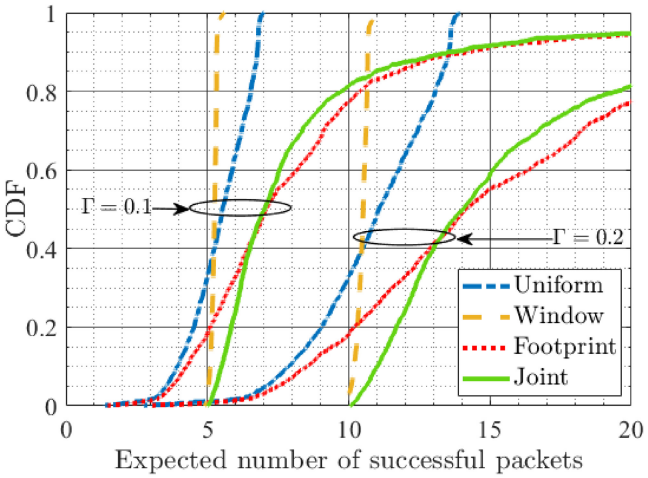


FIGURE 6. CDF of the terminals' expected throughput for $N = 1200$, $\Gamma = 0.1$ (120 slots per minute) and 0.2 (240 slots per minute), or for multi-user cancellation: $\overline{K} = 10$.

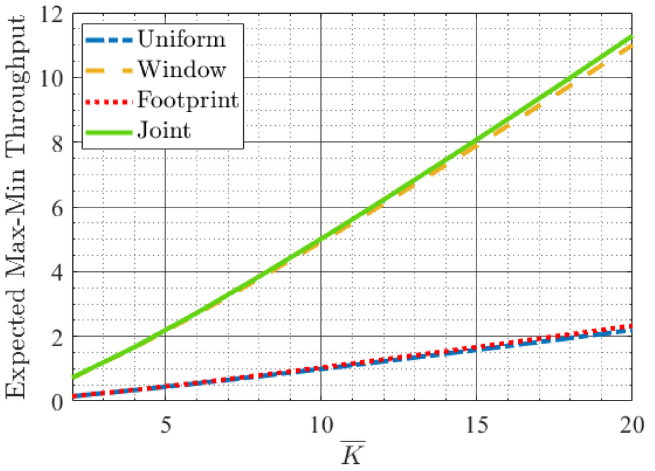


FIGURE 7. Max-min throughput versus the canceller threshold $\overline{K}$.

rate for all terminals, fixed also across their visibility windows. The common attempt rate β is chosen to maximize throughput at the most congested time.

B. THROUGHPUT COMPARISON

In this subsection, we compare the throughput distribution across the terminal population for the four schemes.

Recall from Theorem 2 that the asymptotic success process is inhomogeneous Poisson with intensity $\beta_u(t)\Sigma(H(t))$. Applying this result, the throughput (expected number of successful transmissions) of a terminal at location u is

$$\int_0^T \beta_u(t)\Sigma(H(t))dt \tag{27}$$

Fig. 6 presents the empirical throughput cumulative distribution functions (CDFs) for all four schemes, for a terminal population of $N = 1200$, and for two values of $\Gamma = 0.1$ and $\Gamma = 0.2$ that correspond to 120 slots per minute and 240 slots per minute respectively. The CDFs present the throughput distribution (expected number of successful packets in $[0, T)$) across terminals. Thus for each CDF curve, the value on the Y-axis is the fraction of terminals that achieve a throughput less than or equal to the corresponding X-axis value.

It can be seen from the CDFs for $\Gamma = 0.1$, the Window scheme achieves a throughput of ≈ 5 for all the terminals. Note that this max-min throughput value under the Window scheme is achieved simultaneously for small windowed terminals as well as for terminals with long windows.

Under the Uniform scheme, 40% of the terminals do not achieve this threshold of 5 successes. The worst-case terminal has a throughput of ≈ 1 whereas the best has a throughput of ≈ 7. Under Footprint, 20% of the terminals fall below the threshold of 5 successes, however, the top 6% of the terminals achieve extremely high throughputs (> 20). The Joint scheme has the best overall performance, *i.e.,* it guarantees 5 successes for all terminals without sacrificing the high throughputs of the best placed terminals.

Finally, it can be observed that the CDFs for all the schemes shift to the right as Γ is increased from 0.1 to 0.2. This is because doubling Γ doubles the number of available resources, and leads to a proportional increase in throughputs.

The primary factor in driving capacity of these IoT systems is the effectiveness of the multi-user canceller. We now demonstrate the potential gains and performance improvements achievable by enhancing the multi-user interference cancellation capability.

Fig. 7 presents the improvement in max-min throughput as the canceller threshold ($\overline{K}$) is increased, for the four schemes. As can be seen, the max-min throughput increases almost linearly with $\overline{K}$ for all schemes. The Window and Joint schemes significantly outperform the Footprint and Uniform schemes.

C. SINGLE MESSAGE APPLICATION

In this subsection, we consider the scenario where each terminal has one message to send over the considered time interval $[0, T)$. This message is encoded into packets that will be attempted throughout the duration $[0, T)$. The terminal is successful if and only if one of the transmitted packets is successful.

For this application, we compare the four schemes in terms of the expected fraction of the terminal population that is successful, versus the terminal population size N. We take $\overline{K} = 10$ for multi-user cancellation, and we take $\Gamma = \frac{240}{N}$ (so that there are 240 slots per minute). In this case, we can increase the number of terminals by increasing N, but we also decrease the resources per terminal, since $N\Gamma$ is held fixed at 240 slots per minute. We are therefore increasing the terminal population for the same fixed amount of resources.

Applying Theorem 2, the total number of successful transmissions (denoted $\mathcal{S}_u$) of a terminal at location u is a Poisson random variable with mean $\int_0^T \beta_u(t)\Sigma(H(t))dt$. The message failure probability for a terminal at u equals $\mathbb{P}[\mathcal{S}_u = 0] = e^{-\int_0^T \beta_u(t)\Sigma(H(t))dt}$. Therefore, the message

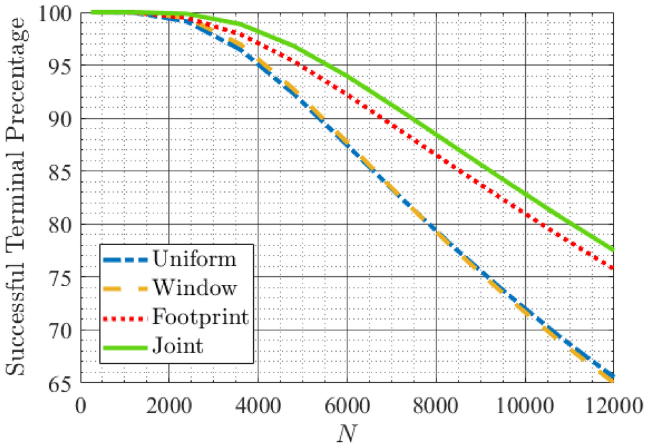


FIGURE 8. Fraction of the population successful with one message per terminal.

success probability is given by

$$1 - e^{-\int_0^T \beta_u(t)\Sigma(H(t))dt} \tag{28}$$

Therefore, the fraction of the successful population is

$$\int_{u\in\mathcal{A}} \left(1 - e^{-\int_0^T \beta_u(t)\Sigma(H(t))dt}\right) \nu(u)du \tag{29}$$

It can be seen from Fig. 8 that all four schemes achieve success for 99% terminals for lightly congested networks $N \leq 2500$ terminals. As the population increases, the percentage of successful terminals decreases due to higher congestion.

The Joint scheme has the best performance of all schemes, followed by the Footprint scheme. The Footprint and Joint schemes exploit the capacity and always transmit at $\Gamma\beta_{\max}$, which has led to superior performance over Window, which load balances across terminal window sizes. Joint has the best performance since it exploits the channel as well as prioritizing terminals with small windows (See also Fig. 6).

D. MULTIPLE MESSAGES APPLICATION

In this subsection, we consider a scenario where terminals generate messages at regular intervals during the visibility duration. Each terminal will only attempt to send the most recently generated message. Although the terminals do not receive feedback on whether each message is successfully received, our framework can be applied to determine the probability of successful delivery of each message.

This is illustrated for four terminal locations: Tasmania, Sydney, Lake Eyre, and Perth (see Fig. 2(a)). A new message is generated by a terminal every 0.5 minute. The visibility durations and the number of messages for each terminal are tabulated in Table 3, for the satellite pass in Fig. 2(a). We take the same Γ, $\overline{K}$ and N as the single message scenario.

Let $\{T_{u,m}\}$ denote the mth message generation time of a terminal at u. Applying Theorem 2 as in the previous subsection, the mth message's success probability is given by

$$1 - e^{-\int_{T_{u,m}}^{T_{u,m+1}} \beta_u(t)\Sigma(H(t))dt} \tag{30}$$

TABLE 3. Terminals visibility duration and the number of messages when the inter-message interval is 0.5 minute.

Terminal Name	Tasmania	Sydney	Lake Eyre	Perth
$\lvert I_u \rvert$ (minutes)	2.1	6.9	10.1	10.2
# messages	4	13	20	20

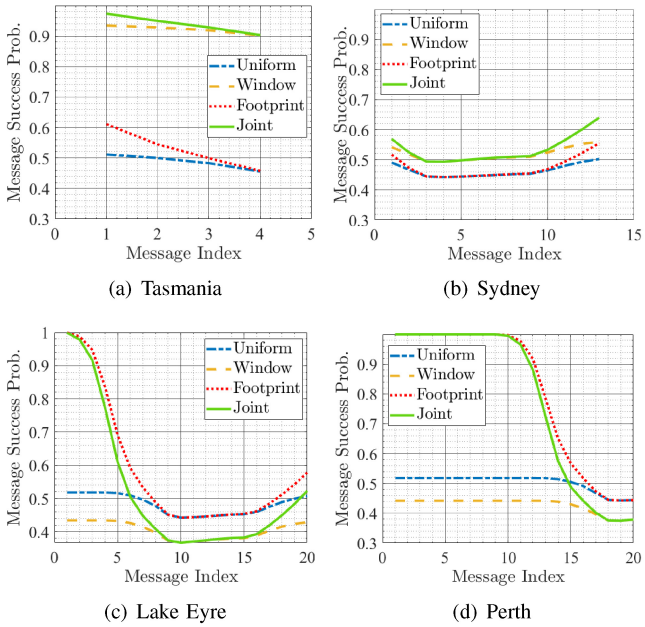


FIGURE 9. Message success probability over the visibility duration for four terminals located at Tasmania, Sydney, Lake Eyre, & Perth.

Fig. 9 illustrates the variation in success probability versus the message index, for the four terminal locations. For the terminal in Tasmania, which has a short visibility window, both the Joint and Window schemes maintain a message success probability above 0.9 for all four messages.

For a short initial duration, Lake Eyre experiences low congestion, resulting in a high success probability (under the Joint and Footprint schemes) for the first few messages. However, as the satellite moves and system congestion increases, the success probability for all schemes drops below 0.5. A similar pattern is observed with the terminal in Perth, where the first 12 messages have a success probability above 0.9, but this drops below 0.5 as more terminals begin to attempt transmission. Despite having similar visibility durations, the congestion profiles for Lake Eyre and Perth are somewhat different. The satellite becomes visible to Perth much earlier than it becomes visible to Lake Eyre. During this time, there is little congestion from other terminals. The satellite is still not very congested when it becomes visible to Lake Eyre, but congestion sets in quickly after that as the satellite becomes visible to many terminals on the Eastern side of Australia.

The following conclusions can be draw from Fig. 9. One is that message success rates can vary considerably across terminals at different geographic locations for all the schemes (Window, Footprint and Joint). Another is that message success rate can vary considerably across time, for any given

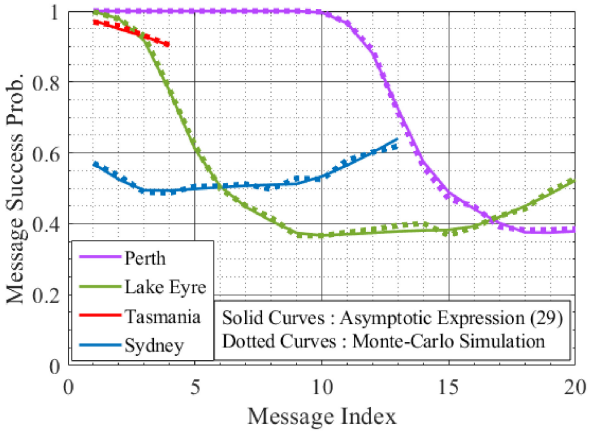


FIGURE 10. Asymptotic expression (30) versus Monte-Carlo simulation for the Joint scheme, for $N = 1200$ and $\Gamma = 0.2$ (240 slots per minute).

geographic location, although the variation across time is much less for Window than the other two schemes.

Uniform and Window do not require any time varying feedback from the satellite, whereas Footprint and Joint require (limited) time varying feedback. The Window scheme's message success rate can be lower than Uniform's, but not much lower, whereas Window's success rate can also be much higher. Joint appears to achieve the best of either Window or Footprint, as one would expect.

Fig. 10 compares the message success probability results in (30), obtained using our asymptotic framework, (solid curves) with Monte-Carlo simulation results (dotted curves). The comparison is shown for the Joint scheme. The figure shows that our asymptotic analytic result is extremely accurate for this practical network size. We have observed the same level of accuracy for the other attempt rate schemes that we have considered in this paper (although not shown here).

In this subsection we have shown that message success probabilities can vary across the terminals even though the message generation at the terminals is regular. We have also shown that the predictions from our Theorem 2 (inhomogeneous Poisson process result) are highly accurate and closely match the Monte-Carlo simulation. We note that the same theorem can be used to analyze the performance of irregular message generation timings, as well as coded transmission schemes.

E. SINGLE MESSAGE APPLICATION WITH A QoS TARGET

1) QoS ACCUMULATION

Consider a terminal in a single message scenario. Let $t_1, \ldots, t_K$ denote the time points in the interval $[0, T)$ at which transmission attempts are made by the terminal. From Theorem 1, the probability of message success at time t_k (*i.e.,* resulting from all the attempts prior to and including t_k) is given by $P_u(t_k) \coloneqq 1 - \Pi_{i=1}^{k}(1 - \Sigma(H(t_i)))$ By denoting $Q_u(t_k) \coloneqq \log_{10}(1 - P_u(t_k))$ to be the message log-failure probability, note that

$$Q_u(t_k) = \sum_{i=1}^{k} \log_{10}(1 - \Sigma(H(t_i))) \tag{31}$$

can be interpreted as an accumulation of QoS over the attempts, where $0 \geq Q_u(t_1) \geq Q_u(t_2), \geq \ldots \geq Q_u(t_k)$.

When the accumulated QoS $Q_u(t_k)$ falls below a target threshold $-\rho$ (where $\rho > 0$), the message failure probability is guaranteed to be lower than $10^{-\rho}$. This construction can be exploited at the terminal to provide reliable message transmission. For example, in a multiple message scenario, a new message can be generated only after sufficient QoS is accumulated for the current message, ensuring that each message is successful with a probability higher than a chosen threshold. For implementing such a scheme, the terminal only requires to know the congestion function values $\{H(t_i)\}_{i=1}^{k}$ at the attempt instances $\{t_i\}_{i=1}^{k}$ to evaluate $Q_u(t_k)$. This information can be obtained from the satellite, using the same feedback mechanism used in the Joint and Footprint schemes.

2) TERMINAL SHUTDOWN SIMULATION

We now focus on a single message scenario where QoS accumulation is used for shutting down terminals. The terminal shuts down after the chosen QoS target threshold is met and, as a result, the congestion is reduced for the remaining active terminals. The reduced congestion benefits the various attempt schemes as follows. In the Window scheme, where the attempt rates are not adapted over time, the common success probability increases due to reduced congestion. In the Footprint and Joint schemes, where the attempt rates are adapted, the remaining terminals are able to increase their attempt rates.

To illustrate the benefit of terminal shutdown and QoS accumulation, we consider two threshold values of 10^{-1} and 10^{-2} and compare with the results in Fig. 8, where terminals do not shut down. The parameter values are $\overline{K} = 10$, $\Gamma = \frac{240}{N}$ (*i.e.,* 240 slots per minute) where N varies from 240 (lightly loaded) to 12000 (heavily loaded).

As seen in Fig. 11, it can be noted that terminal shutdown dramatically improves the performance of the Joint and Footprint schemes at high values of N *i.e.,* in heavily loaded systems. For $N = 9600$ and threshold 10^{-1}, Joint's performance improves from 84% to 96%, whereas Footprint's performance improves from 82% to 94%, which means 12% more terminals achieve success. With the Window scheme, there is only a marginal improvement, which shows that the benefit of boosting the attempt rates (as done in Footprint and Joint) is significant when terminal shutdown is implemented.

Finally, the curves for different threshold values illustrate the trade-off in performance at light loads versus high loads. The joint curve for 10^{-2} achieves 99% terminal success for values of N up to 6000, whereas 10^{-1} achieves only 96% but for values of N up to 9600. This demonstrates

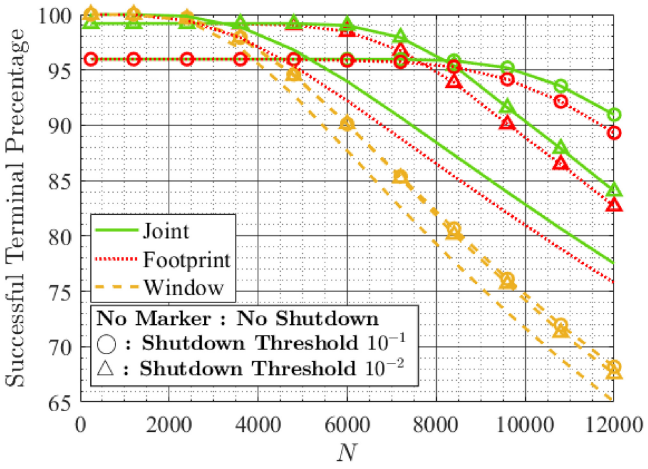


FIGURE 11. Fraction of the successful population in a single message application, for shutdown thresholds 10^{-1} and 10^{-2}.

that if a QoS threshold is satisfactory in the considered application, shutting down by applying QoS accumulation drastically increases supported network size.

VI. CONCLUSION AND FUTURE WORK

A framework has been presented which can be used in the analysis of satellite IoT networks. Our results show that key processes such as terminal attempts are successes are Poisson, in the large population limit. The limit is already attained for a modest sized network with only 240 terminals and a slot duration of 0.5 sec.

We then developed a range of terminal attempt schemes with different implementation complexities. We emphasise that all of these schemes adapt to the changing dynamics of the satellite passes by changing their attempt rates to the dynamic feedback from the satellite. The Footprint and Joint schemes adapt in real-time during a satellite pass to dynamic congestion information broadcast by the satellite. The window scheme is less complex, and adapts only from OB to OB. The performance of the proposed schemes were compared in terms of individual terminal QoS and also population wide QoS *e.g.,* the throughput distribution. Amongst these, the Joint scheme was seen to have the best overall performance.

Furthermore, specific results for a number of IoT applications were obtained. In the QoS accumulation application, it was seen that by using a shutting down mechanism, a significantly larger (12% more) proportion of terminals can be supported. This is another example of how our schemes can adapt to dynamic feedback from the satellite when congestion conditions change over time.

In the paper, the attempt rate schemes were evaluated on the basis of a max-min fair QoS objective. However, the schemes can be readily extended to other objectives, *e.g.,* weighted max-min which leads to differentiated QoS.

We have used the Poisson process limit to obtain results for three IoT applications. Having a process limit is important for analyzing general IoT applications. This is because performance objectives cannot always be captured in terms of marginal events and, instead, must be expressed in terms of sample paths.

APPENDIX A
PROOF OF THEOREM 1

Proof of Theorem 1 (9): It is helpful to begin with the case where the feedback $w(t)$ is a constant w throughout $[0, T]$. The proof will then be generalized to the case of piece-wise constant feedback using the same arguments.

For $t \in [0, T]$, define $\mathcal{S}_{\mathrm{E}}(t) := \bigcup_{t' \in [0,t]} \mathcal{F}(t')$ to be the area swept out by the satellite footprint in $[0, t]$. Correspondingly, let $\mathcal{S}_{\mathrm{L}}(t) := \mathcal{S}_{\mathrm{E}}(t) - \mathcal{F}(t)$ be that area which goes out of the satellite footprint in the same interval $[0, t]$.

In the pre-limit, define

$$E^{(N)}(t) := \int_{u \in \mathcal{S}_{\mathrm{E}}(t)} h(|I_u|, w)\ \nu^{(N)}(du),$$
$$L^{(N)}(t) := \int_{u \in \mathcal{S}_{\mathrm{L}}(t)} h(|I_u|, w)\ \nu^{(N)}(du).$$

Note that $E^{(N)}(t), L^{(N)}(t) \in D^{\uparrow}[0, T]$, *i.e.,* they are monotonically non-decreasing functions with "jumps" (discontinuities of the first kind) on $[0, T]$.

As $h(|I_u|, w)$ is integrable with respect to the measure $\nu(du)$, so that $E(t) := \int_{u \in \mathcal{S}(t)} h(|I_u|, w)\nu(du)$ and $L(t) := \int_{u \in \mathcal{L}(t)} h(|I_u|, w)\nu(du)$ are well defined for $t \in [0, T]$. Moreover, both $E(t), L(t)$ are continuous on $[0, T]$.

The proof for the constant feedback case can now be completed as below. Note that

$$H^{(N)}(t) = \frac{1}{\Gamma}\Big(E^{(N)}(t) - L^{(N)}(t)\Big), t \in [0, T]$$

Hence it is sufficient to show that both $E^{(N)}(t)$ and $L^{(N)}(t)$ converge uniformly to $E(t), L(t)$ respectively, on $[0, T]$ as $H(t) = \frac{1}{\Gamma}(E(t) - L(t))$.

It follows from the Strong Law of Large Numbers that there is pointwise convergence

$$E^{(N)}(t) \to E(t);\ \ L^{(N)}(t) \to L(t),\ \ \forall t \in [0, T]$$

Uniform convergence now follows from [36, Lemma 4.1], since both $E^{(N)}(t), L^{(N)}(t)$ are step functions with bounded steps of $O(1/N)$. It also follows from the above that $H(t)$ is continuous. The proof for constant feedback is now complete.

For the general case, where feedback changes every T_f seconds, $H(t)$ is piecewise continuous. This can be established by applying the above argument for each feedback interval $[iT_f, (i + 1)T_f)$, taking $w = w[i]$. Since we are now working with the semi-open interval $[iT_f, (i + 1)T_f)$, the argument must be modified to assume $w = w[i]$ even at $t = (i + 1)T_f$. This establishes uniform convergence with the modified attempt function defined on $[iT_f, (i + 1)T_f]$. This implies that uniform convergence holds *a fortiori* on the interval $[iT_f, (i + 1)T_f)$ where the actual feedback is $w[i]$. ■

Corollary 1: Suppose $f:(0,\infty)\to[0,1]$ is differentiable with bounded derivative, $|\dot{f}(x)|\le\zeta<\infty,\ \forall\, x$. Then it holds that

$$f(H_u^{N)}(t))\to f(H(t)) \tag{32}$$

uniformly in $[0,T]$ a.s.

This is a consequence of the Mean Value Theorem.

Proof of Theorem 1 (10): Define $\phi_{u,\tau}^{(N)}(z):=\mathbb{E}[z^{C_u^{(N)}(\tau)}]$ and $\mathcal{F}_u^{(N)}(\tau):=\mathcal{F}(\tau)\bigcap\mathcal{U}^{(N)}-\{u\}$. The log-probability generating function $\varphi_{u,\tau}^{(N)}(z):=\log\phi_{u,\tau}^{(N)}(z)$ satisfies

$$\begin{aligned}\varphi_{u,\tau}^{(N)}(z)&=\sum_{u'\in\mathcal{F}_u^{(N)}(\tau)}\log\left(1-\frac{\beta_{u'}(\tau)}{N\Gamma}(1-z)\right)\\&=-\sum_{u'\in\mathcal{F}_u^{(N)}(\tau)}\left(\frac{\beta_{u'}(\tau)}{N\Gamma}(1-z)+\frac{\kappa_{u'}^2}{2N^2\Gamma^2}\right)\end{aligned}$$

where $|\kappa_{u'}|\le 2\beta_{u'}(\tau)\le 2h_{\max}$ by choosing $h_{\max}$ to be the upper-bound on attempt rate function.

Using (8), we may rewrite the above as

$$\varphi_{u,\tau}^{(N)}(z)=-(1-z)\ H^{(N)}(\tau)+\varepsilon_N(\tau,z)$$

where $|\varepsilon_N(\tau,z)|\le\frac{2h_{\max}}{N\Gamma}$.

Let $r_{u,k}^{(N)}(\tau)$ be the coefficient of z^k in the expansion of $\phi_{u,\tau}^{(N)}(z)$. Note that $r_{u,k}^{(N)}(\tau)=\mathbb{P}[C_u^{(N)}(\tau)=k]$ and that the generating function $-\lambda(1-z)$ corresponds to a Poisson distribution with mean λ. Hence, we obtain

$$r_{u,k}^{(N)}(\tau)=\mathcal{P}\left(k;H^{(N)}(\tau)\right)\times(1+O(1/N))$$

uniformly. Hence by Corollary 1, $r_{u,k}^{(N)}(\tau)\to\mathcal{P}(k;H(\tau)),k\in\mathbb{Z}_+$ uniformly. Therefore convergence $\sigma_u^{(N)}(\tau)\to\Sigma(H(\tau))$ (from (3)) is also uniform. ■

APPENDIX B
PROOF OF THEOREM 2

Proof of Theorem 2 (11): We will show that the number of attempts $Y_u^{(N)}(a,b)$ of a terminal at location u in any arbitrary interval $[a,b)\subseteq I_u$ converges to a Poisson random variable with mean $\int_a^b\beta_u(t)dt$. Due to the fact that the attempt processes have independent increments, this will be enough to determine the finite dimensional distributions of the attempt process $A_u^{(N)}(t)$. The weak convergence to the Poisson process now follows from [31, Th. 12.6].

Recall that $\beta_u(t)$ is piece-wise constant. Let $a_0\le a_1\le\cdots\le a_A$ denote points such that $\beta_u(t)$ is constant on the interval $[a_i,a_{i+1})$, where $a_0=a$ and $a_A=b$. For the Nth system, the number of attempts in interval $[a_i,a_{i+1})$ is

$$Y_u^{(N)}(a_i,a_{i+1})=\sum_{j=\left\lfloor\frac{a_i}{T_s^{(N)}}\right\rfloor}^{\left\lfloor\frac{a_{i+1}}{T_s^{(N)}}\right\rfloor}B_i^{(N)}[j]$$

where $B_i^{(N)}[j]$ are i.i.d Bernoulli random variable with mean $\beta_u(a_i)T_s^{(N)}=\frac{\beta_u(a_i)}{N\Gamma}$. Using Poisson law of small numbers [33], $Y^{(N)}(a_i,a_{i+1})$ converges to a Poisson random variable with mean $(a_{i+1}-a_i)\beta_u(a_i)$. Hence, $\sum_{i=0}^{A}Y_u^{(N)}(a,b)$ is a Poisson random variable with mean $\sum_{i=0}^{A-1}(a_{i+1}-a_i)\beta_u(a_i)=\int_a^b\beta_u(t)dt$ since increments are independent, this completes the proof. ■

Proof of Theorem 2 (12): Note that the success process $S_u^{(N)}(t)$ is obtained by an independent marking (or thinning) of the attempt process $A_u^{(N)}(t)$. The marking probability $\sigma_u^{(N)}(t)$. From (11) of Theorem 2, the attempt process of a terminal at location u converges to a Poisson process $\Pi_{\beta_u(t)}$, and from Theorem 1, the marking probability converges to $\Sigma(H(t))$. Hence the process convergence of $S_u^{(N)}(t)$ to $\Pi_{\beta_u(t)\Sigma(H(t))}$ follows the same arguments in the proof of Theorem 2 (11). ■

APPENDIX C
PROOFS OF LEMMA 1, THEOREM 3 AND THEOREM 4

Proof of Lemma 1: Setting $\alpha_k=S_k/k$, we may write $\Sigma(x)=e^{-x}\sum_{k=0}^{\overline{K}-1}\frac{x^k}{k!}\alpha_{k+1}$. Let $M(x):=x\Sigma(x)$ and $f(x):=-\Sigma'(x)$. Also for sake of convenience, let $S_{\overline{K}+1}:=0$ which implies $\alpha_{\overline{K}+1}=0$. Note that

$$f(x)=e^{-x}\left(\sum_{k=0}^{\overline{K}-1}\frac{x^k}{k!}\alpha_{k+1}-\sum_{k=0}^{\overline{K}-2}\frac{x^k}{k!}\alpha_{k+2}\right)$$

with $f(x)>0$ for $x>0$ as $\alpha_k\ge\alpha_{k+1}$ and $\alpha_{\overline{K}}>0$. Also $\Sigma(x)>0$ for $x>0$.

At a stationary point, $M'(x)=\Sigma(x)-xf(x)=0$ or equivalently $\frac{xf(x)}{\Sigma(x)}=1$. We now show that $\frac{xf(x)}{\Sigma(x)}$ is monotonically strictly increasing and therefore that there is a unique stationary point.

Observe that

$$\frac{xf(x)}{\Sigma(x)}=x\left(1-\frac{\sum_{k=0}^{\overline{K}-2}\frac{x^k}{k!}\alpha_{k+2}}{\sum_{k=0}^{\overline{K}-1}\frac{x^k}{k!}\alpha_{k+1}}\right)$$

The ratio of the polynomials on the RHS is monotonically decreasing as the coefficients in the denominator are at least as large as the corresponding coefficients in the numerator. Moreover the denominator has an additional term. Thus $\frac{xf(x)}{\Sigma(x)}$ is monotone strictly increasing and therefore $xf(x)/\Sigma(x)$ equals 1 at a unique point. ■

Proof of Theorem 3: It is straightforward to verify that $\boldsymbol{\beta}^*$ is a feasible solution of LP (13)-(15). We will now prove that it is optimal by showing that w must be less than or equal to w^* for any feasible solution.

From constraint (14), $\beta_u\ge\frac{z}{|I_u|},\forall u\in\mathcal{A}$. Substituting in (15), we obtain $w\int_{u\in\mathcal{F}(t)}\frac{\nu(u)}{|I_u|}du\le\Gamma\beta_{\max},\forall\ t\in[0,T)$. Since $t^*\in[0,T)$, this implies $w\le w^*$. ■

Proof of Theorem 4: From (20), note that for each $t\in[0,T]$, $\beta_t\le\frac{\Gamma\beta_{\max}}{\int_{u\in\mathcal{F}(t)}\nu(u)du}=\frac{\Gamma\beta_{\max}}{V(t)}$. Hence, the feasible solution $\beta_t=\frac{\Gamma\beta_{\max}}{V(t)}$ is also optimal. The optimal objective value w^* is given by $\min_{u\in\mathcal{A}}\int_{t\in I_u}\frac{\Gamma\beta_{\max}}{V(t)}dt$. ■

APPENDIX D
SOLUTIONS TO THE TWO CLUSTER NUMERICAL EXAMPLE

A.GIS SCHEME

1)CASE 1: $\beta_B^{(C)} > 0$ OR EQUIVALENTLY $\frac{\alpha(\eta+1)}{\eta} > 1$

Note that if $\beta_B^{(c)} > 0$, then both constraints in (26) must be tight, as otherwise the objective value could be increased. Hence in this case, by tightness of constraints in (26), we obtain

$$\beta_B^{(c)}|I_A| + \beta_B^{(nc)}(|I_B| - |I_A|) = \beta_A|I_A|$$

This combined with (25) provide a system of linear equations with three variables, which when solved provides the optimal LP solution as follows:

$$\beta_B^{(nc)} = \frac{\Gamma\beta_{\max}}{\nu_B} = \Gamma\beta_{\max}(1+\eta)$$
$$\beta_B^{(c)} = \frac{\Gamma\beta_{\max}}{\alpha}(\alpha(\eta+1)-\eta)$$
$$\beta_A = \frac{\Gamma\beta_{\max}}{\alpha}$$

It can now be observed that $\beta_B^{(c)} > 0$ occurs if and only if $\frac{\alpha(\eta+1)}{\eta} > 1$.

The throughputs for terminals can now be obtained as

$$R_A^{(glo)} = \beta_A\Sigma(\beta_{\max})|I_A| = \overline{R}$$
$$R_B^{(glo)} = \left(\beta_B^{c}|I_A| + \beta_B^{(nc)}(|I_B| - |I_A|)\right)\Sigma(\beta_{\max}) = \overline{R}$$

2)CASE 2: $\beta_B^{(C)} = 0$ OR EQUIVALENTLY $\frac{\alpha(\eta+1)}{\eta} \leq 1$

Note that if $\beta_B^{(c)} = 0$, from (25), we obtain the solution of the LP as

$$\beta_A = \frac{\Gamma\beta_{\max}}{\nu_A} = \Gamma\beta_{\max}\left(1+\frac{1}{\eta}\right)$$
$$\beta_B^{(c)} = 0;\ \beta_B^{(nc)} = \frac{\Gamma\beta_{\max}}{\nu_B} = \Gamma\beta_{\max}(1+\eta)$$

Hence, the throughput for a cluster A terminal is

$$R_A^{(glo)} = \beta_A|I_A|\Sigma(\beta_{\max}) = \overline{R}\alpha\left(1+\frac{1}{\eta}\right)$$

whereas for a cluster B terminal, it is

$$R_B^{(glo)} = \left(\beta_B^{(c)}|I_A| + \beta_B^{(nc)}(|I_B| - |I_A|)\right)\Sigma(\beta_{\max})$$
$$= \overline{R}(1-\alpha)(1+\eta)$$

B.FOOTPRINT SCHEME

In the Footprint scheme, during the congestion period, the common terminal attempt rate is $\Gamma\beta_{\max}$ (*i.e.,* $\beta_A = \beta_B^{(c)} = \Gamma\beta_{\max}$) whereas during non-congestion, it is $\frac{\Gamma\beta_{\max}}{\nu_B} = \Gamma\beta_{\max}(1+\eta)$ (*i.e.,* $\beta_B^{(nc)} = \Gamma\beta_{\max}(1+\eta)$).

Hence for the terminals in cluster A, the throughput is given as

$$R_A^{(fpt)} = \Gamma\beta_{\max}\Sigma(\beta_{\max})|I_A| = \overline{R}\alpha$$

whereas for terminals in cluster B, it is given by

$$R_B^{(fpt)} = \Gamma\beta_{\max}\Sigma(\beta_{\max})|I_A| + \Gamma\beta_{\max}(1+\eta)\Sigma(\beta_{\max})(|I_B| - |I_A|)$$
$$= \overline{R}(1+\eta-\alpha\eta)$$

C.WINDOW SCHEME

In the Window scheme, let $\beta_A = \frac{w}{|I_A|}$ and $\beta_B^{(c)} = \beta_B^{(nc)} = \frac{w}{|I_B|}$ be the attempt rates of terminals in cluster A and cluster B respectively.

During the bottleneck congestion period, the aggregate per-slot attempt rate $\frac{1}{\Gamma}\left(\beta_A\nu_A + \beta_B^{(c)}\nu_B\right)$ equals $\beta_{\max}$. Hence we obtain

$$w = \frac{\Gamma\beta_{\max}}{\frac{\nu_A}{|I_A|} + \frac{\nu_B}{|I_B|}} = \frac{\Gamma\beta_{\max}|I_B|(1+\eta)}{\frac{\eta}{\alpha}+1} \tag{33}$$

Note that w is the expected number of total attempts for any terminal.

The throughput for a cluster A terminal is given as follows:

$$R_A^{(win)} = w\Sigma(\beta_{\max}) = \overline{R}\frac{\alpha(\eta+1)}{\eta+\alpha}$$

whereas for terminals in cluster B, it is

$$R_B^{(win)} = \beta_B|I_A|\Sigma(\beta_{\max}) + \beta_B(|I_B| - |I_A|)\Sigma\left(\frac{\nu_B\beta_B}{\Gamma}\right)$$
$$= \overline{R}\frac{\alpha(\eta+1)}{\eta+\alpha}\left(\alpha + (1-\alpha)\frac{\Sigma\left(\frac{\beta_{\max}\alpha}{\eta+\alpha}\right)}{\Sigma(\beta_{\max})}\right)$$

Since $\Sigma(x)$ is monotonically decreasing,

$$\left(\alpha + (1-\alpha)\frac{\Sigma\left(\frac{\beta_{\max}\alpha}{\eta+\alpha}\right)}{\Sigma(\beta_{\max})}\right) \geq 1$$

Hence, $R_B^{(win)}$ is slightly larger than $R_A^{(win)}$.

D.JOINT SCHEME

Joint scheme behaves the same as the window scheme during congestion. Hence, $\beta_A = \frac{w^{(c)}}{|I_A|}$ and $\beta_B^{(c)} = \frac{w^{(c)}}{|I_B|}$ are attempt rates during congestion where

$$w^{(c)} = \frac{\Gamma\beta_{\max}|I_B|(1+\eta)}{\frac{\eta}{\alpha}+1}$$

from (33) and moreover

$$R_A^{(jnt)} = R_A^{(win)} = \overline{R}\frac{\alpha(\eta+1)}{\eta+\alpha}$$

For cluster B terminals, during non-congestion, the attempt rates are scaled up, and are given by

$$\beta_B^{(nc)} = \frac{\Gamma\beta_{\max}}{\nu_B} = \Gamma\beta_{\max}(1+\eta)$$

Hence, the throughput for cluster B terminals is given as

$$R_B^{(\text{jnt})} = \beta_B^{(\text{c})}|I_A|\Sigma(\beta_{\max}) + \beta_B^{(\text{nc})}(|I_B| - |I_A|)\Sigma(\beta_{\max})$$
$$= \overline{R}\left(\frac{\alpha^2(\eta+1)}{\eta+\alpha} + (1-\alpha)(1+\eta)\right)$$
$$= \overline{R}(1+\eta)\left(1 - \frac{\eta\alpha}{\eta+\alpha}\right)$$

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